

Kartikey Pathak

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EDUCATION

Veermata Jijabai Technological Institute (VJTI)

B.Tech in Electrical Engineering, Minor in Robotics (CGPA: 8.04)

Mumbai, India

2023 – 2027

EXPERIENCE

Eyecandy Robotics

Electronics and Hardware Engineer

April 2026 – August 2026

Bangalore, India

- Designed production PCBs for a legged consumer robot: Raspberry Pi Zero 2W and Milk-V compute carriers, servo driver boards, and a power board running 17 motors off one pack
- Owned schematic capture, routing and fabrication output, integrating IMU, servo bus, audio and microphone array with test points for hardware bring-up

NishCorp Technologies

Robotics Control Systems Engineer Intern

July 2025 – November 2025

Hybrid

- Developed Gazebo simulation for a robotic wheelchair platform to analyze motion and steering, applying differential drive kinematics and integrating IMU plugins
- Contributed to the half-scale hardware prototype through integration, testing and field evaluation

Krishna Defence and Allied Industries Limited

Research Intern - Defence Technology

April 2025 – July 2025

IIT Gandhinagar

- Developed firmware for Qorvo DWM3001CDK Ultra-Wideband modules using Zephyr RTOS for indoor positioning where GPS is unreliable, handling low-level drivers and RF packets
- Implemented SLAM and NAV2 for autonomous forklift POC navigation in warehouse environments
- Achieved precision localization error of 30 cm across a 10,000 sq meter area and 2 cm in a 50 sq meter area

PROJECTS

PCB Axial-Flux BLDC Motor | KiCAD, FEMM, Python

Aug 2026 – Present

- Designing a coreless axial-flux motor whose stator is the PCB itself: 100 mm two-layer board, 6 wedge coils over 8 poles, dual rotor with 16 N42 magnets across a 0.5 mm air gap
- Built FEMM magnetostatic models to extract torque constant, back-EMF and torque ripple, identifying a 3.3x torque loss from the plastic rotor before fabrication

FOC Controller for Micro BLDC Motor | STM32, DRV8311H, PCB Design

Jan 2026 – Feb 2026

- Developed a Field-Oriented Control (FOC) system for a 4300 KV micro BLDC motor operating from a 3.7 V supply
- Designed and routed a custom PCB integrating an STM32 microcontroller, DRV8311H motor driver, and AS5600 encoder

Smart Switch Board | ESP32, ESP-IDF, MQTT, BLE, Flutter, PCB Design

2025 – Present

- Built a network-controlled mains wall switch across every layer: CAD enclosure, a relay and optocoupler board isolating 220 V from a 3.3 V ESP32, ESP-IDF firmware, a TLS MQTT broker and a Flutter app
- Kept the physical switches working with network, broker and app unavailable, using BLE WiFi provisioning and state persisted in NVS across power cuts

Open Manipulator X Simulation | ROS2, Gazebo, RViz, Computer Vision

June 2024 – July 2024

- Programmed forward and inverse kinematics for an Open Manipulator X robot arm to perform precise pick-and-place tasks
- Implemented object detection and tracking algorithms to locate target objects in the Gazebo simulation environment

COMPETITIONS

eYantra Warehouse Drone Competition | IIT Bombay

Oct 2024 – Feb 2025

- Developed autonomous navigation and PID control algorithms for stable hovering and motion of a simulated drone, with ArUco marker detection for localization and task execution

POSITIONS OF RESPONSIBILITY

Society of Robotics and Automation (SRA)

Mechanical Head & Active Member

Aug 2024 – Mar 2026

VJTI, Mumbai

- Conducted technical workshops on Embedded C, ESP32 microcontrollers, and ROS-based systems for 200+ students
- Mentored junior teams in developing self-balancing and line-following robots, providing hardware/software expertise

TECHNICAL SKILLS

Programming Languages: C, C++, Python, Embedded C

Software & Frameworks: ROS2, Gazebo, RViz, ESP-IDF, STM32CubeIDE, KiCAD, micro-ROS

Hardware: ESP32, STM32, Raspberry Pi, UWB Modules, Motor Drivers, BLDC/Stepper/DC Motors, LiDAR

Core Competencies: Robot Kinematics, Field-Oriented Control, SLAM, NAV2, PID Control, PCB Design, IoT